Naveen Harshitha Karunanayake

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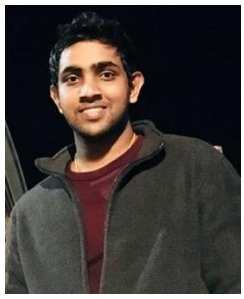

PROFILE

I am carrying myself as a dynamic individual with a brilliant academic career and a passion for learning new concepts, technologies, and looking for a career with emerging and challenging ventures. Hence, I have equipped myself with skills in various aspects and my solutions incorporate creativity, innovation and sustainability for a problem at hand. I strongly believe that I am competent at effective communication and creative thinking.

EDUCATION

- ***2015-2020: B.Sc. Engineering (Hons)*** *specialized in Electronic and Telecommunication Engineering from University of Moratuwa, Sri Lanka.*

  Overall GPA – 3.91/4.20 (First Class Honors)

- ***2001-2014: G.C.E. Advanced Level Examination, G.C.E. Ordinary Level Examination, Grade 5 scholarship Examination,*** *Rahula College, Matara, Sri Lanka.*

  GCE Advanced Level Examination 2014 – Physical Science

  District Rank: 1

  National Rank: 19 ***Z-Score: 2.7479*** High distinction for all 4 subjects

  GCE Ordinary Level Examination 2011

  District Rank: 5 High distinction for all 9 subjects

WORK EXPERIENCE

- ***Machine Learning Engineer and Project Manager : F Code Labs (PVT) LTD , Sri lanka (April 2020 – Present)***
- ***Graduate Research Assistant : University of Moratuwa, Sri lanka (March 2020 – June 2020)***
- ***Research Intern : Data61|CSIRO Sydney, Australia (June 2018 – December 2018)***
  - ✓ A Multi-modal Neural Embeddings Approach for Detecting Mobile Counterfeit Apps
  - ✓ Logo synthesis using Generative Adversarial Networks

PUBLICATIONS

- **Naveen Karunanayake**, Jathushan Rajasegaran, Ashanie Gunathillake, Suranga Seneviratne, and Guillaume Jourjon, "**A Multi-modal Neural Embeddings Approach for Detecting Mobile Counterfeit Apps: A Case Study on Google Play Store**," in **IEEE Transactions on Mobile Computing**, doi: 10.1109/TMC.2020.3007260. *Paper*
- Jathushan Rajasegaran, **Naveen Karunanayake**, Ashanie Gunathillake, Suranga Seneviratne, and Guillaume Jourjon. **A Multi-modal Neural Embeddings Approach for Detecting Mobile Counterfeit Apps**. In Proceedings of the 2019 **World Wide Web Conference (WWW '19)**, May 13–17, 2019, San Francisco, CA, USA. ACM, New York, NY, USA, 7 pages. *Paper Forbes Article*
- **Naveen Karunanayake**, Achintha Wijesinghe, Chameera Wijethunga, Chinthani Kumaradasa, Peshala Jayasekara, and Ranga Rodrigo, **"Towards a Smart Opponent for Board Games: Learning beyond Simulations"** in Proceedings of **IEEE International Conference on Systems, Man, and Cybernetics** (**IEEE SMC 2020**), October 11-14, 2020, Toronto, CA.
- Sicong Wang, **Naveen Karunanayake,** Tham Nguyen, and Suranga Seneviratne. **''Privacy Preserving Spam Filtering using Functional Encryption''-** submitted to **IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks** (**IEEE WOWMOM 2021**). *Paper* - Under Review

ACHIEVEMENTS AND AWARDS

- **All-Island 19th (Ranked 1st in the District) in GCE Advanced Level Examination 2014** held by the Government of Sri Lanka
- **Ranked 5th in the District in GCE Ordinary Level Examination 2011** held by the Government of Sri Lanka
- **The Best Student Award in Mathematics in the year of 2014 –** Awarded by Rahula College, Sri Lanka
- **Dialog Merit Scholarship Award 2014 –** For Outstanding performance in GCE Advanced Level 2014
- **Dialog Merit Scholarship Award 2011 –** For Outstanding performance in GCE Ordinary Level 2011
- **Runners Up –** IEEE IAS Robotics Demonstration contest 2020
- **Bronze Award – InnovateFPGA 2018 –** Asia Pacific Region
- **2nd Runners Up – Code Experts 3.0** – Algorithmic coding competition organized by University of Moratuwa
- **6th place (Out of 80 teams)** – **MoraXtreme 1.0 Coding Competition** organized by University of Moratuwa
- **IEEExtreme 10.0 - International Algorithmic Coding Competition organized by IEEE**
  Island Rank - 6 | World Rank - 181
- **Awarded Mahapola Higher Education Merit Scholarship for** achieving excellence in the GCE A/L Examination

ONLINE COURSES AND CERTIFICATIONS

- **Machine Learning** by Stanford University on Coursera –Certificate earned in August 2017
- **Neural Networks and Deep Learning** by deeplearning.ai on Coursera – Certificate earned in January 2018
- **Convolutional Neural Networks** by deeplearning.ai on Coursera – Certificate earned in April 2018
- **Structuring Machine Learning Projects** by deeplearning.ai on Coursera – Certificate earned in April 2018
- **Improving Deep Neural Networks: Hyper parameter tuning, Regularization and Optimization** by deeplearning.ai on Coursera – Certificate earned in April 2018
- **Reinforcement Learning Specialization** by University of Alberta on Coursera. (Ongoing)

PROJECT EXPERIENCES

- **Deep Reinforcement Learning Based Carrom Manipulator – Final Year Project**
  An end to end system which continuously captures the board configuration using computer vision and learns to play the best possible shots to complete a game using Reinforcement learning techniques.
  i) Computer Vision based piece detection and localization.
  ii) Rule-based algorithm development to extract parameters of the best possible shots.
  iii) Unity based carrom simulator to predict the configuration after playing a shot.
  iv) Training the carrom agent to learn the best possible shots through Deep Reinforcement learning techniques.
- **Application of Neural Embeddings for detecting mobile counterfeit apps in google play store and analysis for malware, dangerous permissions and third party ad libraries – Research Project (Data61- CSIRO)**
  i) Generating neural embeddings for 1.2M app data in Google play store,
     a. To represent content and style features using convolutional neural networks.
     b. To represent text features using Word2Vec/Doc2Vec models.
  ii) Retrieving potential counterfeits for the most downloaded apps in Google Play store.
  iii) Analysing potential counterfeits for malware, dangerous android permissions and third party ad libraries.
- **Plant Disease recognition – Group Project**
  i) Combination of multiple spectral images (NIR/RGB) to extract features.
  ii) Training of a convolutional neural network for classification.
- **Proximity Meter using Micro Controllers - Group Project**
  A device to measure the distance to an object using IR and Sonar sensors.
- **Analog Line Follower – Group Project**
  A Line following robot using Operational Amplifiers (Without using any Micro Controllers) which also consists of PID controlling circuits.
- **Automated GPS based Navigation Robot – Group Project**
  Simple robot for reaching a given destination while avoiding obstacles, following walls. Integration of multiple sensors in real time.
- **Smart Protector to avoid Overvoltage failure**
- **Buck Converter**

- **Design Project -Simulation of a Band Pass filter using Kaiser Window**
- **Implementing UART using FPGA**

TECHNICAL AND PERSONAL SKILLS

- **Programming Languages –** PYTHON (Keras, Tensorflow, Scikit-learn, NLTK, OpenCV, Pandas), MATLAB, ARDUINO PROGRAMMING, JAVA, Adaptable to any programming language
- **General Languages –**Fluent in both English and Sinhala – Written and Spoken
- **Software –**Wing IDE, Net Beans IDE, ORCAD, SolidWorks, Mikro C, Microsoft Office and General Application Software(adaptable to any software)
- Fast learner, Excellent problem solving skills, Good communication skills, Team Leadership, Team Management, Team player and Motivated

ACADEMIC AND RESEARCH INTERESTS

- **Machine Learning and Deep Learning**
- **Big Data Analysis**
- **Computer Vision**
- **Cyber Security**
- **Algorithm Development and Optimization**
- **Image Processing**

EXTRA CURRICULAR ACTIVITIES

- **CHESS –** Won School Colours and Participated in Provincial and All-Island competitions
- **SCRABBLE –**Ranked 8th in the Provincial Tournament in 2010
- **BADMINTON –**Member of the School Badminton squad

SOCIETIES AND LEADERSHIP ACTIVITIES

- **Senior Prefect –** Rahula College
- **Team Leader –** Science Team, Mathematics Team, Scrabble Team of Rahula College
- **Organizing Committee Member –** EXMO 2017 National level exhibition organized by University of Moratuwa
- **Treasurer –** IEEE Industrial Application Society chapter, University of Moratuwa
- **Active Member –** Electronic Club of University of Moratuwa
- **Student Member –** IEEE Organization
- **Volunteer –** IEEE Student Branch of University of Moratuwa

REFERENCES

1. Dr. Suranga Seneviratne
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2. Dr. Ranga Rodrigo
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